

12.3 Predefined Streams

Besides the file pointers which we explicitly open by calling `fopen`, there are also three predefined streams. `stdin` is a constant file pointer corresponding to standard input, and `stdout` is a constant file pointer corresponding to standard output. Both of these can be used anywhere a file pointer is called for; for example, `getchar()` is the same as `getc(stdin)` and `putchar(c)` is the same as `putc(c, stdout)`. The third predefined stream is `stderr`. Like `stdout`, `stderr` is typically connected to the screen by default. The difference is that `stderr` is not redirected when the standard output is redirected. For example, under Unix or MS-DOS, when you invoke

```
program > filename
```

anything printed to `stdout` is redirected to the file `filename`, but anything printed to `stderr` still goes to the screen. The intent behind `stderr` is that it is the "standard error output"; error messages printed to it will not disappear into an output file. For example, a more realistic way to print an error message when a file can't be opened would be

```
if((ifp = fopen(filename, "r")) == NULL)
{
    fprintf(stderr, "can't open file %s\n", filename);
    exit or return
}
```

where `filename` is a string variable indicating the file name to be opened. Not only is the error message printed to `stderr`, but it is also more informative in that it mentions the name of the file that couldn't be opened. (We'll see another example in the next chapter.)

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